

5.1

$$q_{ii} = - \left. \frac{d}{dh} P_{ii}(h) \right|_{h=0} = \lambda_i$$

$$q_{ij} = \left. \frac{d}{dh} P_{ij}(h) \right|_{h=0} = \begin{cases} \lambda_i, & j=i+1 \\ 0, & j=i-1 \end{cases}$$

$$\Rightarrow Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & & & \\ & -\lambda_1 & \lambda_1 & & \\ & & -\lambda_2 & \lambda_2 & \\ & & & \dots & \\ & & & & -\lambda_n & \lambda_n \end{pmatrix}$$

① Kolmogorov 向前方程: $P'(t) = P(t)Q$

$$P'_{ij}(t) = P_{i(i-1)}(t) \lambda_{j-1} - P_{ij}(t) \lambda_j$$

② Kolmogorov 向后方程: $P'(t) = Q P(t)$

$$P'_{ij}(t) = -\lambda_i P_{ij}(t) + \lambda_i P_{(i+1)j}(t)$$

$$\text{由 ①} \quad \begin{cases} P_{ii}(t) = e^{-\lambda_i t} \\ P_{ij}(t) = e^{-\lambda_j t} \int_0^t e^{\lambda_j s} \lambda_{j-1} P_{i,i-1}(s) ds, \quad j \geq i+1 \end{cases}$$

5.2

$$P_{ij}(h) = \begin{cases} \frac{1}{2}h + o(h), & i \neq j \\ 1 - h + o(h), & i = j \end{cases}$$

$$q_{ij} = \frac{1}{2}, \quad i \neq j \quad q_{ii} = -1$$

$$Q = \begin{pmatrix} -1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -1 \end{pmatrix}$$

向前 Kolmogorov 方程: $P'(t) = P(t)Q$

$$\text{① } P'_{ij}(t) = -P_{ij}(t) + \frac{1}{2}P_{i,j-1}(t) + \frac{1}{2}P_{i,j+1}(t)$$

$$\lambda p_{i,j-1}(t) + \mu p_{i,j+1}(t) = (1 - p_{ij}(t))$$

$$\textcircled{1} \Rightarrow p'_{ij}(t) = \frac{1}{3} - \frac{2}{3} p_{ij}(t)$$

$$\text{解得 } p_{ij}(t) = c e^{-\frac{2}{3}t} + \frac{1}{3}$$

$$\text{当 } p_{ij}(0) = 1, i=j, c = \frac{2}{3}$$

$$p_{ij}(0) = 0, i \neq j, c = -\frac{1}{3}$$

$$\therefore p_{ij}(t) = \begin{cases} \frac{1}{3} - \frac{1}{3} e^{-\frac{2}{3}t}, & i \neq j \\ \frac{1}{3} + \frac{2}{3} e^{-\frac{2}{3}t}, & i = j \end{cases}$$

$$\text{平稳分布 } \pi_j = \lim_{t \rightarrow \infty} p_{ij}(t) = \frac{1}{3}, j = 1, 2, 3$$

5.3

(1)

$$p_{ij}(h) = \begin{cases} i\mu h + o(h), & j = i-1 \\ (M-i)\lambda h + o(h), & j = i+1 \\ 1 - (\lambda_i + \mu_i)h + o(h), & j = i \\ o(h), & |j-i| \geq 2 \end{cases}$$

$$\text{令 } \lambda_i = (M-i)\lambda, \mu_i = i\mu$$

Kolmogorov 向前方程:

$$p'_{i,j}(t) = p_{i,j-1}(t)\lambda_{j-1} - p_{i,j}(t)(\lambda_j + \mu_j) + p_{i,j+1}(t)\mu_{j+1}$$

$$i, j \in I$$

$$\text{讨论平稳分布 } \begin{cases} \pi_j q_{ij} = \sum_{i \neq j} \pi_i q_{ij} \\ \pi_0 q_{00} = \pi_1 q_{10} \end{cases}$$

$$\Rightarrow \begin{cases} \lambda_0 z_0 = \mu_1 z_1 \\ z_i (\lambda_i + \mu_i) = z_{i-1} \lambda_{i-1} + \mu_{i+1} z_{i+1}, \end{cases}$$

$$\Rightarrow \begin{cases} z_0 = \left(1 + \sum_{j=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{j-1}}{\mu_1 \mu_2 \cdots \mu_j} \right)^{-1} \quad j \geq 1 \\ z_j = \frac{\lambda_0 \lambda_1 \cdots \lambda_{j-1}}{\mu_1 \mu_2 \cdots \mu_j} \left(1 + \sum_{j=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{j-1}}{\mu_1 \mu_2 \cdots \mu_j} \right)^{-1}, \end{cases}$$

$$\begin{aligned} (2) \quad P(N(t) > 5) &= \sum_{j=6}^{10} z_j = \sum_{j=6}^{10} C_{10}^j \left(\frac{60}{90}\right)^j \left(\frac{30}{90}\right)^{10-j} \quad (j \geq 1) \\ &= 0.7809 \end{aligned}$$

5.4

$$p_{ij}(h) = \begin{cases} \mu h + o(h), & j = i-1 \\ \lambda h + o(h), & j = i+1 \\ 1 - (\lambda + \mu)h + o(h), & j = i \\ o(h), & |j-i| \geq 2 \end{cases}$$

$$Q = \begin{pmatrix} -\lambda & \lambda & & \\ \mu & -(\lambda+\mu) & \lambda & \\ & \mu & -(\mu+\lambda) & \lambda \\ & & \mu & -\mu \end{pmatrix}$$

Kolmogorov 向前方程组
(定理 5.7)

$$\begin{cases} P_0'(t) = -\lambda P_0(t) + \mu P_1(t), \\ P_1'(t) = \lambda P_0(t) - (\lambda + \mu) P_1(t) + \mu P_2(t), \\ P_2'(t) = \lambda P_1(t) - (\lambda + \mu) P_2(t) + \mu P_3(t), \\ P_3'(t) = \lambda P_2(t) - \mu P_3(t) \end{cases}$$

初值条件 $p_0(0)=1, p_j(0)=0, j=1,2,3$

对 $\forall i \in I, \lim_{t \rightarrow \infty} p_{ij}(t) = \pi_j = p_j$

J. 5

略, 题目答案有问题

$$f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots$$

5. b

服务员服务时间 T 服从指数分布:

$$f(t) = \begin{cases} \mu e^{-\mu t}, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

(1)

$$E[E(x(t)|T=t)] = \int_0^{\infty} \left(\sum_{n=0}^{\infty} n \frac{(\lambda t)^n}{n!} e^{-\lambda t} \right) \mu e^{-\mu t} dt$$

$$\text{其中, } \sum_{n=0}^{\infty} n \frac{(\lambda t)^n}{n!} e^{-\lambda t} = e^{-\lambda t} \sum_{n=0}^{\infty} n \frac{(\lambda t)^n}{n!} = e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda t)^n}{(n-1)!}$$

$$= \lambda t \cdot e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1}}{(n-1)!} \quad \text{Taylor 展开 } e^{\lambda t}$$

$$= \lambda t \cdot e^{-\lambda t} \cdot e^{\lambda t} = \lambda t$$

$$\therefore \text{原式} = \int_0^{\infty} \lambda t \cdot \mu e^{-\mu t} dt = \frac{\lambda}{\mu}$$

(2)

$$p_0 = \int_0^{\infty} e^{-\lambda t} \cdot \mu e^{-\mu t} dt = \frac{\mu}{\mu + \lambda}$$